

Ecological adaptation

In the environment living organism tackle every situation created by the nature in order to grow properly and give continuity to the generation. So, in order to gain maximum benefit from the resources around the nature the organism develops certain features in them for the sustainability of their generation. The features may be physical, anatomical, morphological, reproductive or any other changes.

That's why in short **we can say that adaptation is a process of adapting to the nature by the certain changes in function of organism by which an organism becomes better suited to its environment.**

Types of Adaptation

classification on the basis of their water requirements and nature of habitat which includes the following three types of adaptations:

1. **Hydrophytic Adaptation**
2. **Mesophytic Adaptation**
3. **Xerophytic Adaptation**

Here, we mainly shade the light in related with hydrophytic Adaptation and Xerophytic Adaptation. So, before going towards the topic we have to know about some terms and terminologies with their meaning which will help you a lot in dealing with above mentioned adaptations.

Aids to memory

- **Lithophytes:** Growing on the surface of rocks
- **Halophytes:** Growing on saline soil
- **Psammophytes:** Growing on the sand
- **Oxylophytes:** Growing on acid soils
- **Sclerophytes:** Growing on forest
- **Hydrophytes:** Hydrophytes are the plants that grow in wet places and the plant get either partly or wholly submerged in water. For eg: Lotus, Pistia, Hydrilla, Wolffia etc.

Hydrophytes can be further classified into 3 types on relation with water and air which are discussed below:

Floating hydrophytes: They are the plants that float on the surface of the water which have contact between both the water and air. They are further divided into two parts which are mentioned below:

i. Floating rooted hydrophytes: In this category the submerged plants are rooted in muddy substratum but their leaves and other shoots system including flowering one float on the water surface. These plants have wax layers. For eg: Water lily, Lotus etc.

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ii. Free floating hydrophytes: In this case the plants are not rooted in the muddy substrata but the plants float freely on the surface of the water. For eg: *Salvinia*, *Wolffia*, *Azolla* etc.



=(Azolla)

2.Submerged hydrophytes: Those plants which are not in contact with soil and grow below the water surface is called submerged hydrophytes. For eg: *Hydrilla*, *Chara*,



Utricularia etc.

=(Hydrilla)

3.Amphibian hydrophytes: Those plants which are rooted in the muddy substrata and shoot system are exposed to the air is commonly known as amphibian hydrophytes. For eg: *Neptuma*, *Commelina*, *Phragmites* etc



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=(Phragmites)

The aquatic plants have the following adaptations:

1. Roots are completely absent (*Salvinia*, *Azolla* etc.) and some had poorly developed roots like in *Hydrilla*.
2. The plants have fibrous root which are mainly unbranched.
3. In submerged hydrophytes the stem is long, weak slender and flexible but short, thick and spongy in case of free-floating plants. But in some cases, stems are modified into rhizome or runner.
4. The leaves of free-floating hydrophytes are green, smooth , shining, long and thin which are covered with the smooth waxy surface. Here, the wax coating protects the leaves from chemical and physical injuries.
5. Pollination and seed dispersal take place through the water.
6. The cuticle is totally absent in submerged parts, but it may be present in aerial part as thin layer in hydrophytes.

7. The stomata is completely absent in submerged parts but in some cases vestigial stomata is present. Here gaseous exchange takes place through cell wall. In floating plants stomata are confined in upper surface.
8. In submerged plants the aerenchyma is well developed and air chamber in aerenchyma are filled with respiratory moisture and gases.
9. In hydrophytes the vascular elements like xylem and phloem are poorly developed and vascular bundles are arranged at the center.
10. In hydrophytes the mechanical tissue are poorly developed or absent and the thick-walled sclerenchymatous tissue is completely absent in both the floating and submerged hydrophytes.

Xerophytes: Those plants which grow in dry conditions are called xerophytes. For e.g.: *Cycas*, *Opuntia*, *pinus* etc.

Xerophytes are classified into further two categories based on drought resisting power and on the basis of their morphology and physiology.

On the basis of drought resisting power : Based on drought resisting power, xerophytes are again divided into two groups which are mentioned below:

i. **Drought i.Resistant plants:** Those plants which can resist extreme droughts by developing certain adaptive feature are called drought resistant plants. They are mostly found in deserts, gravel and sand.

ii. **Drought escaping plants:** They are mostly found in semi arid zones(zone where rainy season area of short time period) . They survive as seeds and fruits during dry time and during favorable conditions the seed germinate into new plants. So in this way they remain unaffected by the extreme conditions, that's why they are called drought evaders and ephemeral.

On the basis of their morphology and physiology: On the basis of their morphology and physiology , xerophytes are classified into 3 types which are discussed below:

i. Succulents: They are drought resisting xerophytes which grow in the place where there is no water or water found in less amount. They have the fleshy leaves which stores the large amount of water whenever available. For eg: *Opuntia*, *Cactus*, *Aloe* etc.



=(Opuntia)

ii. Ephemeral ii. Annuals: Those xerophytes which completes their life cycle before the arrival of dry condition and mostly found in arid zones in annual basis are called ephemeral annuals. For eg: *Argemone mexicana*, *Solanum xanthocarpum*, *Cassia tora* etc.



=(Solanum xanthocarpum)

iii. True xerophytes (non-succulent xerophytes): Those xerophytes which grow in dry condition and cannot store the water are called true xerophytes. For eg: *Calotropis procera*, *Acacia nelotica*, *Alhagi* etc.

=(Calotropis procera)

The xerophytic plants have the following adaptations:

- 1. The root system is well developed with the presence of root hairs which helps in the absorption of the water.**
- 2. The stem of xerophytes are mainly hard and woody. In some xerophytes stem are modified into thorns, cladode and phylloclades.**
- 3. In xerophytes mainly the leaves are reduced to scales and if the leaves are present then they are caducous.**

- 4. In some xerophytes needle shaped leaves are present in order to reduce the rate of transpiration. For eg: Pinus**
- 5. Xerophytes has thick cuticle on leaves and waxy layer .**
- 6. Fruits and seeds are covered by hard shells and flowers develop in suitable condition.**
- 7. In xerophytic plants the mechanical and conducting tissue are well developed.**
- 8. Sunken stomata is present which help in the reduction of transpiration rate.**
- 9. In some xerophytes epidermal hairs are present in the stem of leaves to protect stomata and prevent the excessive water loss.**
- 10. In some plants aerenchyma may be present for the storage of water.**